MIL-M-45690A(MU)
18 NOVEMBER 1963

SUPERSEDING
MIL-M-4S690(ORD)
30 NOVEMBER 1960

MILITARY SPECIFICATION

MOUNT, TELESCOPE: M114

1. SCOPE

1.1 This specification covers one type of Telescope Mount known as Mount, Telescope: M114.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in efFect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MIL-F-13926 - Fire Control Materiel: General Specification Governing the Manufacture and Inspection of

MIL-P-14232 —Parts, Equipment and Tools for Ordnance Materiel: Packaging of

MIL-1-45208 —Inspection requirements, General Specification for

STANDARDS

MILITaRY

M1L-STD-105 —Sampling Procedures and Tables; For Inspection by Attributes

MIL-STD-109 —Quality Assurance Twins and Definitions

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

F8574690 —Mount, Telescope: M114

LSQAP 8574690—List of Supplementary Quality Assurance Provisions for Mount, Telescope: M114

IEL 8574690 --List of Inspection Equipment Numbers for Mount, Telescope: M114

PACKAGING AND DATA SHEET

MIL-P-14232 /P8574690-Packaging of Mount, Telescope: M114

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication. The mount shall be manufactured in accordance with Drawing F8574690 and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to insure inherent quality and performance capabilities.

3.3 General specification. The contractor shall be responsible for adherence to, and compliance with the following requirements of Speification MIL-F-13926:

FSC 1240

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(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.4 Environmental.

3.4.1 Vibration. With the mount secured in its normal operating position and both boresight movements secured by means of the locking levers, the mount shaIl be vibrated at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes plus or minus 15 seconds, after which it shall meet all requirements of this specification.

3.4.2 Temperature. The mount shall be exposed to ambient temperatures of minus 40 degrees and plus 150 degrees F. until thermal equilibrium is reached at each temperature and inspected at each temperature to the applicable requirements of 3.4.2.1.

3.4.2.1 Operability. The running torque required to rotate the azimuth and Elevation Boresight knobs shall not exceed 8 inch pounds when checked at room temperature 60° to 90° F., and shall not exceed 15 inch pounds when checked with the mount thermally stabilized at -40° F. and at +150°F. The slip scales shall be movable at any of the above temperatures.

3.4.3 Bellows. Bellows shall be examined co insure against deterioration after the mount has been subjected to the temperatures of 3.4.2.

3.5 Performance.

3.5.1 Alinement. By means of the azimuth boresight knob, the holder assembly shall be set at its midpoint of travel in azimuth, and the scale slipped to the (3) position in coincidence with the index and, the elevation boresight knob set to the extreme limit in depression, and the scale slipped to the "I" position o determine the following requirements.

3.5.2 Movement. Theazimuth knob shall provide a minimum of 20 mils movement in each direction and the elevation knob shall provide a minimum of 40 mils in elevation as read on the respective scale, bucklash excluded.

3.5.2.1 Backlash. Backlash in the elevation and azimuth boresight movement shall not exceed 0.05 mil,

3.5.3 Boresight locks. With a 60 pound inch load applied to each boresight lock, there shall be no movement in azimuth or elevation when a torque of 20 pound inches is applied to each boresight knob. Locking with 60 pounds inches torque and unlocking of the boresight locks shall not cause any change in the azimuth and elevation alinement of the mount.

3.5.4 Ho/der assembly. The holder assembly shall permit installation and clamping of a medium having a spherical mounting surface measuring 3.106 inches in diameter. The assembly shall remain rigid; removal shall be without restriction, and the clamping screw shall not be adversely affected as a result of having clamped the spherical surface in the holder assembly and applying a force of 6 pounds parallel to the longitudinal axis of the spherical surface.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or my other inspection facilities and services acceptable to the Government, Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves thc right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.l.1 General provisions. The quality aSsurrance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by

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the supplier. Inspection shall be in accordance with the requirements of Specification MIL-I-45208 and the component inspection requirements of specification MIL-F-13926. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Inspection lot. Unless oterwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Presentation of Product" specified in Standard MIL-STD-105.

4.2.2 Examination and tests. Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL-F-13926 and List of Supplementary Quality Assurance Provisions, LSQAP-8574690. Examinations and tests related to section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL-STD-105, and the inspection level and sampling p]ans specified in table I titled "classification of Defects." Examinations and tests for Packaging, Packing and Marking shall be in accordance with Specification MIL-P-14232 and section 5 herein. The tabulated classificfition of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual nOncOnforming items.

TABLE I. Classification of defects
Use Level II of table I with Sampling Plan table II-A of Standard M1L-STD-105.

	Requirement	Test procedure
Critical: None.		
Major: AQL 1.0%		
101. Vibration	3.4.1	4.5.1
102. Alinement	3.5.1	4.5.4
103. Movement	3.5.2	4.5.5
104. Backlash	3.5.2.1	4.5.6
105. Boresight locks	3.5.3	4.5.7
106. Holder assembly	3.5.4	4.5.8
Minor: AQL 2.5%		
201. Bellows	3.4.3	Visual

4.2.3 Disposition of nonconforming product. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance or Rejection" as specified in Standard MIL-STD-105.

4.3 Control tests.

4.3.1 General. One mount shall be selected at random from each 100 produced, or from each months production, Whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

Control test	Requirement	Test procedure
Materials	3.1	Per drawings & Specifications
Fabrication	3.2	Visual & Drawings
General construction	3.3	MIL-F-13926

4.3.2 Temperature tests. Three mounts shall be selected at random from each 50 produced, or from each months production, whichever occurs first, to determine compliance with operational temperatures as specified in 3.4.2, 3.4.2.1, 4.5.2 and 4.5.3.

4.3.3 Control test failure. Should the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment. Except as otherwise provided for by the contract, the contractor Shallfurnish and maintain all required measuring and testing equipment in accordance with the "Inspection Equipment" requirement of Specification MIL-I-45208. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Gorernment furnished inspection equipment. Where the contract provides for Government furnished test equipment, care

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and maintenance of test equipment shall be in accordance with "Inspection Equipment," and "Government Furnished Material" requirements of Specification MIL-1-45208.

4.4.2 Contractor furnished inspection equipment.

4.4.2.1 GOVERNment design. All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied by the contractor in accordance with the List of Inspection Equipment Numbers, IEP8574690. Alternate designs may be substituted only as provided by "Inspection Provisions" of Specification MIL-I-45208.

4.4.2.2 Contractor design, The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL-F-18926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will pertnit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90 % of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment,

4.5 Test methods and procedures.

4.5.1 vibration. Vibration shall be in accordance with the Set-up and Operating instructions on Vibration Fixture Drawing F8636537 and the Requirements of 3.4.1. The use of a simulated telescope as shown on Drawing F8636549 shall be used in this test.

4.5.1.1 Post vibration. Subsequent to vibration the mount shall be thorougly examined for physical defects resulting from poor workmanship and materials, prior to being subjected to the remaining performance tests. Minimum visual examination shall include the housing, knobs, locking levers, scales, screws, etc. A tactile examination should also be made to determine any excessive looseness or improper function of knobs, scales, locking levers, etc. Anyone of the possible defects defined above shall be cause to consider a unit mount as defective. Mounts which have passed the above examinations shall be tested for all the requirements of this specification.

4.5.2 Temperature. The mount shall be placed in a test chamber that is capable of maintaining the temperatures as specified in 3.4.2 until thermal equilibrium is reached, While at these temperatures the mount shall be subjected to the tests of 4.5.3. After removing the mount from the test chamber a visual-tactile test as described in 4.5.1.1 shall be made to ascertain that materials and workmanship were not damaged by the extreme temperatures.

4.5.3 Operating knobs and slip scales. The elevation and deflection knobs shall be checked at both the extreme temperatures and at room temperature to the torque values specified in 3.4.2.1 using the adapters and torque wrench as specified in the operating instructions on Drawing F8636560. A tactile examination of the slip scales shall be made at the extreme temperatures while performing the above check on the operating knobs. Subsequent to these checks, the mount shall be examined as defined in 4,5.2.

4.5.4 Alinement. Alinement shall be in accordance with the "set-up instructions" on Drawing F8636560 and the requirements of 3.5,1 to perform the following tests.

4.5.5 Movement. This test shall be performed using Final Inspection Fixture F8636660 and in accordance with the operating instructions therein. Azimuth and elevation knob movement shall be as specified in 3.5.2.

4.5.6 Backlash. The test maybe performed in conjunction with 4.5.5 and the operating instructions of Drawing F8636560. Backlash in elevation or azimuth movement shall not exceed the requirements of 3.5.2.1.

4.5.7 Boresight locks. Apply the load as

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specified in 3.5.3 to each boresight lock in accordance with the operating instructions using the test equipment as specified on final inspection fixture Drawing F8636560. There shall be no deflection of the indicators when applying the load.

4.5.8 Holder assembly. Simulated tele-Scope F8636549 shall be installed into the holder assemb;y, and the required force shall be applied as specified in 3.5.4 in accordance with the operating instruct istructions shown in Drawing F8636560.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Pakaging. packing and marking shall be in accordance with Specification MIL-P-14232 and Packaging Data Sheet MIL-P-l4232/ P8574690. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The Mount, Telescope: MI14 is intended for usc in the 90mm Gun, Tank, M48 series to support the Telescope M105 and in the 105mm Gun Tank M60 and M60AI series to support the Telescopes M105C and M105D.

6.2 ordering data. Procurement documents should specify the following:

(a) Title, number, and date 01 this specification.

(b) Selections of applicable levels of preservation, packaging, and packing. (5.1)

(c) Applicable stock number.

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